

Dynamical Modeling of Energy Poverty

1 Conceptualization of Energy Poverty WorldWide

Energy poverty has increasingly been recognized as a multidimensional and dynamic phenomenon rather than a condition reducible to a single indicator of income or energy expenditure. In general terms, energy poverty refers to a situation in which households are unable to secure adequate energy services necessary for a decent standard of living, including heating, cooling, lighting, cooking, hot water, and the operation of essential household appliances. Such deprivation may result from insufficient income, high energy prices, low energy efficiency of buildings, inadequate infrastructure, or combinations of these factors.

Contemporary research emphasizes that energy poverty extends beyond affordability alone and also includes dimensions of accessibility, reliability, cleanliness, comfort, and the capability to transform energy access into actual wellbeing. In this sense, energy poverty represents not only material deprivation but also a broader limitation of social participation, health, and quality of life.

1.1 Global Perspectives on Energy Poverty

The international literature demonstrates substantial geographical variation in the conceptualization and measurement of energy poverty. In high-income economies, research has traditionally focused on affordability-based indicators, particularly expenditure ratios such as the well-known “10% rule” [?, ?]. However, recent studies criticize fixed expenditure thresholds because they may fail to identify “hidden energy poverty,” where households deliberately suppress energy consumption to maintain low expenditures despite inadequate thermal comfort and compromised wellbeing [?, ?].

In contrast, studies conducted in developing regions, including Sub-Saharan Africa, South Asia, and Latin America, typically adopt multidimensional approaches that incorporate access to modern energy infrastructure, reliability, and energy quality. A widely used framework is the Multidimensional Energy Poverty Index (EPI) [?, ?, ?]. Since physical electrification alone does not guaranty affordable or usable energy services, the EPI evaluates several dimensions of deprivation simultaneously. For households j , the index is commonly defined as

$$EPI_j = \sum_{i=1}^d w_i I_{ij}, \quad (1)$$

where d denotes the number of considered indicators, w_i represents the weight assigned to indicator i , and I_{ij} is a binary deprivation indicator for household j ; see [?]. Current methodological debates mainly concern the specification of the weights w_i , with recent contributions increasingly employing Principal Component Analysis (PCA) instead of equal weighting schemes to reflect regional statistical variability more accurately.

Transitional economies such as China increasingly combine affordability and infrastructural perspectives. Recent studies highlight the importance of clean cooking fuels, winter heating affordability, and environmental sustainability as rural electrification reaches saturation levels [?, ?].

1.2 Social and Demographic Dimensions

Energy poverty interacts strongly with existing social and demographic vulnerabilities. The literature consistently identifies elderly populations, women, rural households, and low-income groups as particularly exposed.

Older adults are especially vulnerable due to fixed incomes, greater thermoregulatory needs, and increased health sensitivity to inadequate housing conditions. Longitudinal studies associate prolonged exposure to cold indoor environments and biomass pollution with deteriorating physical and mental health, including cognitive decline [?, ?, ?].

In many developing regions, energy poverty also exhibits a pronounced gender dimension. Dependence on traditional biomass fuels often generates severe “time poverty,” as women spend substantial amounts of time collecting fuel and managing household energy needs. This restricts educational and economic opportunities while simultaneously increasing exposure to harmful indoor air pollution [?, ?, ?]. Natural disasters and environmental shocks may further intensify these vulnerabilities by damaging infrastructure and forcing households to revert to polluting traditional fuels.

1.3 Dynamic and Temporal Nature of Energy Poverty

Recent research increasingly emphasizes that energy poverty is not a static condition but a dynamic process characterized by persistence, transitions, and vulnerability to external shocks. Households may experience chronic energy poverty, temporary deprivation, or repeated transitions between vulnerable and non-vulnerable states.

Budría and Velasco-Gallego [?] identify several distinct trajectories of energy poverty, including persistently non-poor households, temporarily vulnerable households, volatile households, and long-term declining households. Similarly, Masciandaro et al. [?] show that the probability of entering or remaining in energy poverty differs substantially across countries and household types. Chen et al. [?] distinguish between chronic and transient forms of energy poverty, highlighting the importance of duration and persistence in both measurement and policy design.

This temporal dimension suggests that energy poverty should not be conceptualized solely as a short-term inability to pay energy bills. Instead, it should also reflect structural vulnerabilities that generate long-term risks of deprivation. These vulnerabilities may include unstable income, poor housing quality, demographic pressures, financial exclusion, regional inequalities, and exposure to geopolitical or environmental shocks [?, ?].

1.4 Policy Perspectives and Structural Interventions

The current policy-oriented literature increasingly advocates structural rather than purely reactive approaches to energy poverty alleviation. Direct subsidies for energy bills are often criticized as short-term measures that fail to address underlying structural causes such as low building efficiency and poor housing conditions [?, ?].

Consequently, many studies emphasize long-term interventions, including residential retrofits, energy-efficient renovations, modernization of heating systems, and improved financial inclusion [?, ?]. Digital financial tools and broader access to credit may improve households’ ability to invest in cleaner and more efficient energy technologies.

At the same time, the literature warns about unintended policy consequences. For example, strict environmental regulations without adequate compensatory mechanisms may deepen energy vulnerability among low-income rural populations [?, ?]. Effective policy design, therefore, requires integrated approaches that combine environmental objectives, social protection, housing policy, and infrastructure development [?].

2 Motivation for a Dynamical Perspective and Graduality of Energy Poverty

Energy poverty exhibits substantial persistence and delayed responses to economic, infrastructural, and policy-related shocks. However, the development of empirically validated dynamical models is often constrained by limited longitudinal data availability. In many contexts, sufficiently long and reliable time series do not yet exist, making purely data-driven long-term forecasting difficult.

Nevertheless, the absence of long historical datasets does not invalidate a dynamical approach. A theoretical dynamical framework remains valuable because it formalizes the structural relationships among income, energy expenditures, arrears, housing efficiency, and external policy interventions. Such models make it possible to distinguish between short-term responses, long-run equilibria, transitional dynamics, and feedback effects that cannot easily be captured by static statistical approaches.

An additional advantage of the proposed framework is its analytical tractability. Because the system admits explicit mathematical analysis, one can directly study equilibria, stability properties, adjustment speeds, and parameter sensitivities without relying exclusively on extensive empirical calibration. The primary purpose of the present framework is therefore not immediate forecasting accuracy, but rather the construction of a coherent theoretical benchmark that clarifies the mechanisms underlying energy poverty dynamics and provides a basis for future empirical refinement.

2.1 Indicators for Energy Poverty Dynamics

Within the European Union, energy poverty is generally understood as a situation in which households are forced to reduce energy consumption to a level that negatively affects health and wellbeing. According to the European Commission, the principal drivers of energy poverty include low household income, high energy prices, and poor energy performance of buildings and appliances.

The European approach also recognizes the growing importance of “summer energy poverty,” referring to the inability to maintain safe indoor temperatures during heat waves. This issue has become increasingly relevant in the context of climate change and urban overheating.

Following the COVID-19 crisis, rising energy prices, and geopolitical instability associated with the Russian invasion of Ukraine, energy poverty became a central policy concern across Europe. In response, the EU introduced multiple instruments, including the Social Climate Fund, revised directives on energy efficiency and electricity markets, and the Energy Poverty Advisory Hub, which provides technical assistance and methodological support to local authorities.

As an example, let us mention that in 2025, the Czech Ministry of Industry and Trade formally adopted a national definition of energy poverty following consultations with an expert working group composed of representatives from academia, public administration, and other stakeholders, see [?, ?]. It is operationalized using household-level indicators available in the EU-SILC database. In this framework, energy poverty is conceptualized as a multidimensional condition reflecting insufficient access to socially necessary energy services at an affordable cost, and it employs three complementary indicators:

- I1:** inability to adequately heat or cool the dwelling,
- I2:** arrears on utility or energy payments,
- I3:** simultaneous occurrence of high energy expenditure burden and income poverty.

These indicators capture complementary dimensions of energy vulnerability, including subjective thermal deprivation, financial stress, and objective expenditure burden.

The use of these three indicators is therefore not intended to reduce energy poverty to three isolated symptoms, but to provide a parsimonious operational representation of its main mechanisms within the dynamical system. Each indicator corresponds to a different channel through which energy poverty evolves: inadequate heating reflects unmet energy needs, arrears capture the accumulation of financial pressure, and high energy burden represents the structural imbalance between household resources and energy costs. Together, they allow the model to distinguish between deprivation, debt formation, and affordability constraints. This is important because households may enter energy poverty through different pathways, and the same prevalence level may correspond to different underlying dynamics.

At the same time, the proposed framework is not tied exclusively to the Czech operationalization. In another national context, the three indicators may be replaced by locally relevant measures, such as lack of access to clean cooking fuels, unreliable electricity supply, inability to cool the dwelling, or dependence on traditional biomass. The structure of the dynamical system remains unchanged, while the empirical meaning of its components is adapted to the institutional, climatic, and socio-economic conditions of the country under study. Thus, the model should be understood as a transferable framework: national indicators define how energy poverty is observed, whereas the dynamical system describes how its main dimensions interact and evolve over time.

2.2 From Binary Classification to Gradual Energy Poverty Indices

A substantial limitation of many traditional approaches to energy poverty measurement is the reliance on binary classification schemes. In such frameworks, households are typically classified as either energy poor or non-energy poor according to fixed thresholds. For example, whether energy expenditures exceed a predefined proportion of income or whether a household reports an inability to adequately heat the dwelling; e.g., if I_1, I_2, I_3 is defined as binary, then usually energy poverty is defined as $I_1(x) \vee I_2(x) \vee I_3(x)$ for a household x , hence, also binary. Although these indicators are easy to interpret and operationalize, they introduce an artificial discontinuity into a phenomenon that is inherently gradual and multidimensional.

In reality, energy vulnerability rarely emerges abruptly at a single threshold value. Households located just below and just above a predefined cutoff are often socio-economically very similar, yet binary indicators assign them completely different statuses. This may lead to loss of information, reduced sensitivity to moderate vulnerability, and instability of empirical conclusions under small variations in income, prices, or weather conditions. Furthermore, binary indicators are unable to adequately capture partial deprivation, cumulative disadvantage, or varying intensities of exposure across households.

For these reasons, recent research increasingly emphasizes the importance of gradual and continuous representations of energy poverty. Fuzzy logic provides a natural mathematical framework for such an extension because it allows households to belong to the set of energy-poor households only to a certain degree rather than absolutely. Instead of binary deprivation indicators taking values only in $\{0, 1\}$, fuzzy indicators take values in the interval $[0, 1]$, where higher values represent higher intensity of deprivation.

Under this approach, each household x is characterized by fuzzy indicator degrees

$$\tilde{I}_i(x) \in [0, 1],$$

where $\tilde{I}_i(x)$ expresses the degree to which household x is deprived in dimension i . The multidimensional energy poverty index can then be generalized to the gradual form as

$$FEPI(x) = \bigoplus_{i=1}^d \tilde{I}_i(x), \quad (2)$$

and possibly with weights expressing importance of the particular indicator

$$FEPI_w(x) = \bigoplus_{i=1}^d (w_i \otimes \tilde{I}_i(x)), \quad (3)$$

where $\oplus$ is a generalization of classical disjunction $\vee$, $\otimes$ is a generalization of classical conjunction $\wedge$ and $\tilde{I}_i(x)$ represents the degree of deprivation instead of a binary indicator. The classical binary formulation becomes a special case obtained when all $\tilde{I}_i(x)$ attain only values 0 or 1.

This gradual formulation provides several advantages. First, it preserves substantially more information about the intensity and structure of household vulnerability. Second, it reduces instability caused by arbitrary threshold effects. Third, it enables smoother integration with dynamical models, where abrupt discontinuities may create unrealistic transitions and analytical difficulties. In particular, continuous fuzzy indicators are more suitable for studying persistence, trajectories, and nonlinear feedback mechanisms in energy poverty dynamics. Finally, fuzzy formulations better reflect the social-science interpretation of deprivation as a continuum rather than a strictly dichotomous state.

Consequently, introducing fuzzy logic into multidimensional energy poverty measurement represents not merely a technical modification, but a conceptual shift from categorical classification toward gradual vulnerability assessment. Such an approach is especially relevant when energy poverty is analyzed as a dynamic and evolving process influenced by interacting economic, infrastructural, demographic, and environmental factors.

Note: weights can be present as well.

2.3 Gradual Energy Poverty in a Dynamical Framework

The transition from binary to gradual representations of energy poverty is particularly important when energy poverty is studied as a dynamical phenomenon. Binary indicators are suitable for static classification, but they become problematic in dynamical systems because they introduce abrupt discontinuities into the evolution of household states. In reality, however, households typically move into and out of energy poverty gradually through cumulative economic pressure, changing energy prices, deteriorating housing conditions, or progressive indebtedness. A continuous representation is therefore substantially more consistent with the temporal and structural nature of energy vulnerability.

The incorporation of fuzzy logic into the dynamical framework allows energy poverty to be modeled as a continuously evolving degree of vulnerability rather than as a discrete state transition. Instead of assuming that households instantaneously switch from “non-poor” to “energy poor” after crossing an arbitrary threshold, fuzzy indicators permit smooth transitions reflecting varying intensities of deprivation. This is especially relevant for households located near conventional cut-off values, where small fluctuations in income, weather conditions, or energy expenditures may otherwise produce unrealistic jumps between states.

Within this framework, the dynamical system no longer describes only the evolution of observable economic variables, but also the gradual propagation of deprivation across multiple dimensions. For example, increasing energy prices may initially increase the energy burden component, which subsequently contributes to arrears accumulation and eventually to inadequate heating conditions. Similarly, investments into housing renovation may progressively reduce deprivation intensity instead of causing an instantaneous transition from energy poverty to non-poverty.

An important advantage of this formulation is that the resulting trajectories become smoother and analytically more tractable. Continuous fuzzy indicators naturally integrate with differential or difference equation systems and allow the analysis of persistence, stability, feedback loops, and tipping effects without introducing unrealistic discontinuities. Moreover, the framework permits the modeling of partial policy effects. For instance, a subsidy or renovation program may reduce

the degree of vulnerability even if it does not fully eliminate energy poverty according to a binary definition.

The gradual dynamical perspective also improves comparability across countries and institutional settings. Different national operationalizations may employ different deprivation indicators, thresholds, or weighting schemes, while the underlying dynamical structure remains conceptually unchanged. The model therefore separates two levels of analysis: the empirical specification of deprivation dimensions and the general mechanisms governing their interaction and temporal evolution.

From this perspective, energy poverty should not be interpreted as a fixed categorical status, but rather as a continuously evolving state of household vulnerability generated by interacting economic, infrastructural, demographic, and environmental processes. The integration of fuzzy logic with dynamical systems thus provides a mathematically flexible and socially realistic framework for studying the emergence, persistence, and mitigation of energy poverty over time.

3 Title at work: Structure of Paper

4 Methods used in modeling energy poverty

Energy poverty has emerged as a major barrier to global sustainable development, representing the inability of households to access adequate, reliable, and affordable energy for essential daily needs such as heating, cooling, and lighting. To comprehensively understand and address the multifaceted nature of energy poverty, the academic literature has increasingly deployed a highly diverse and sophisticated range of methodological approaches. These methods span from multi-dimensional index construction and causal econometric modeling to advanced spatial analyses, artificial intelligence, and systems thinking.

5 Dynamical Representation Based Only on EU–SILC Variables

5.1 Introduction to the Model

The following model is designed to describe the dynamics of the main observable dimensions entering the definition of energy poverty, rather than the dynamics of the composite energy-poverty indicator itself. In particular, it focuses on the temporal evolution of inability to heat the dwelling adequately, arrears on utility bills, energy burden, and poverty status. These quantities are interpreted as continuous intensity variables derived from EU–SILC indicators.

This approach is appropriate because the composite energy-poverty status is defined only as a logical combination of these dimensions, whereas the dimensions themselves may display persistence and mutual interactions over time. The purpose of the model is therefore to capture how changes in one dimension may affect the others and how adverse conditions may accumulate or gradually diminish. The resulting framework provides a simplified-form dynamical representation of the observable components underlying energy poverty. If the model is required to rely only on variables directly observed in EU–SILC or constructed from them, then a simplified-form dynamical representation is more appropriate than the mechanistic system based on latent technical quantities. In that case, the state is defined directly in terms of the observable dimensions of energy poverty.

Let

$$\dot{x}(t) = f(x(t)), \quad x(t) = \begin{bmatrix} h(t) \\ d(t) \\ b(t) \\ p(t) \end{bmatrix} \in [0, 1]^4,$$

Category	Methods and References
Measurement & Indices	10% Rule, LIHC (affordability thresholds) [?, ?, ?, ?, ?, ?] MEPI (Alkire–Foster multidimensional index) [?, ?, ?] PCA-based weighting (data-driven index construction) [?, ?, ?, ?, ?, ?] FGT-type indices (structural vs transient poverty) [?, ?] Hidden energy poverty (self-rationing) [?, ?]
Econometric Models	TWFE and IV models (causal estimation) [?, ?, ?] DID approaches (policy evaluation) [?, ?, ?, ?] Policy and financial inclusion studies [?, ?, ?, ?, ?]
Spatial Analysis	ESTDA, SDM, GWR (spatial heterogeneity and spillovers) [?, ?] Infrastructure and regional development (SEZ, micro-grids) [?, ?]
Machine Learning	DTW, 1D-CNN (dynamic trajectories) [?] Clustering (household vulnerability profiles) [?, ?, ?] Demographic/aging analysis [?, ?, ?]
Systems Modeling	System Dynamics (feedback systems) [?] GSEM and causal mediation [?, ?] Health and cognitive impacts [?, ?]

Table 1: Overview of methodological approaches in energy poverty research.

Symbol	Meaning
$h(t)$	Convenient temperature
$b(t)$	Energy burden
$d(t)$	Energy-related debts
$p(t)$	Financial poverty
I_1, I_2, I_3	Binary EU–SILC indicators
$\tilde{I}_1, \tilde{I}_2, \tilde{I}_3$	Fuzzy indicators
b_L, b_U	Fuzzy burden thresholds
$\otimes$	Product t-norm: $x \otimes y = xy$
$\oplus$	T-conorm (probabilistic sum)
$\Psi_{0 \rightarrow 1}$	Increasing cosine ramp
$\Psi_{1 \rightarrow 0}$	Decreasing cosine ramp

Table 2: Overview of symbols in use.

where $h(t)$ denotes inability to heat or cool the dwelling adequately, $d(t)$ arrears on utility bills, $b(t)$ the energy-burden component, and $p(t)$ the financial poverty component. All these quantities are either directly observed in EU–SILC or they can be constructed from EU–SILC variables.

State-space invariance. The simplified system is defined on the compact state space $[0, 1]^4$. Under nonnegative coefficients and $\lambda_i > 0$, the hypercube $[0, 1]^4$ is forward invariant. That is, if $x(0) \in [0, 1]^4$, then $x(t) \in [0, 1]^4$ for all $t \geq 0$, provided that trajectories are interpreted as intensities truncated at the boundaries.

This guarantees that the model respects the natural bounds of indicator-based variables and prevents economically meaningless states.

A simple continuous-time simplified system can be written as

$$\dot{h}(t) = \alpha_1 b(t) + \beta_1 p(t) - \lambda_1 h(t), \quad (4)$$

$$\dot{d}(t) = \alpha_2 b(t) + \beta_2 p(t) - \lambda_2 d(t), \quad (5)$$

$$\dot{b}(t) = \alpha_3 b(t) + \beta_3 p(t) + \gamma_3 d(t) - \lambda_3 b(t), \quad (6)$$

$$\dot{p}(t) = \alpha_4 p(t) + \beta_4 d(t) + \gamma_4 h(t) - \lambda_4 p(t), \quad (7)$$

where all coefficients are nonnegative. This system represents mutual reinforcement between the observable dimensions of deprivation while avoiding the introduction of latent variables such as technical efficiency or physical energy need.

Or equivalently, a simple continuous-time simplified system may be formulated as

$$\dot{h}(t) = -\lambda_h h(t) + \alpha_{hb} b(t) + \alpha_{hp} p(t), \quad (8)$$

$$\dot{d}(t) = -\lambda_d d(t) + \alpha_{db} b(t) + \alpha_{dp} p(t), \quad (9)$$

$$\dot{b}(t) = -\lambda_b b(t) + \alpha_{bd} d(t) + \alpha_{bp} p(t), \quad (10)$$

$$\dot{p}(t) = -\lambda_p p(t) + \alpha_{pd} d(t) + \alpha_{ph} h(t), \quad (11)$$

where each diagonal term $-\lambda_i x_i(t)$ captures the tendency of the corresponding dimension to decrease in the absence of reinforcing factors, and the off-diagonal terms represent cross-effects between the observed dimensions of deprivation.

5.2 Interpretation and Justification of the Simplified Dynamic Representation

The simplified system

$$\dot{h}(t) = -\lambda_h h(t) + \alpha_{hb} b(t) + \alpha_{hp} p(t), \quad (12)$$

$$\dot{d}(t) = -\lambda_d d(t) + \alpha_{db} b(t) + \alpha_{dp} p(t), \quad (13)$$

$$\dot{b}(t) = -\lambda_b b(t) + \alpha_{bd} d(t) + \alpha_{bp} p(t), \quad (14)$$

$$\dot{p}(t) = -\lambda_p p(t) + \alpha_{pd} d(t) + \alpha_{ph} h(t) \quad (15)$$

is intended as a simplified-form dynamical representation of the main observable dimensions of energy poverty. Here, $h(t)$ denotes inability to heat the dwelling adequately, $d(t)$ arrears on utility bills, $b(t)$ the energy-burden component, and $p(t)$ the poverty component. All four variables are interpreted either as binary indicators smoothed to the interval $[0, 1]$, or as continuous intensity scores derived from EU-SILC variables.

The diagonal terms $-\lambda_h h(t)$, $-\lambda_d d(t)$, $-\lambda_b b(t)$, and $-\lambda_p p(t)$, with $\lambda_h, \lambda_d, \lambda_b, \lambda_p > 0$, represent the natural tendency of each deprivation dimension to decrease in the absence of reinforcing pressures. In substantive terms, these coefficients capture recovery, adaptation, or exit from the corresponding adverse state. Without such terms, the system would contain only amplification mechanisms and would therefore lack a stabilizing force.

The off-diagonal terms represent cross-effects between the dimensions. In particular, the term $\alpha_{hb} b(t)$ expresses the idea that a high energy burden increases the risk of inadequate heating, since households facing disproportionately high energy costs may reduce consumption below

adequate thermal comfort. The term $\alpha_{hp}p(t)$ reflects the fact that poverty further limits the ability to maintain adequate heating conditions.

Similarly, the equation for $d(t)$ states that arrears are more likely when the household faces a high energy burden or is poor. The coefficient α_{db} captures the mechanism that high energy expenditure relative to available resources increases the probability of falling behind on utility payments, while α_{dp} reflects the broader financial fragility associated with poverty.

The dynamics of $b(t)$ are influenced by $d(t)$ and $p(t)$. The term $\alpha_{bd}d(t)$ expresses that households already in arrears are likely to experience persistent affordability stress, while $\alpha_{bp}p(t)$ captures the empirical fact that poverty is strongly associated with excessive relative energy expenditure.

Finally, the poverty component $p(t)$ is reinforced by both arrears and heating deprivation. The term $\alpha_{pd}d(t)$ reflects the fact that payment difficulties are a manifestation of deeper material hardship, whereas $\alpha_{ph}h(t)$ captures the close link between inability to afford adequate heating and broader poverty or deprivation.

The system is therefore not meant to describe the physical or engineering mechanisms of household energy use. Rather, it is a simplified-form representation of mutual reinforcement among the observed dimensions entering the definition of energy poverty. Its purpose is to capture the qualitative feedback structure:

$$\text{poverty} \longleftrightarrow \text{energy burden} \longleftrightarrow \text{arrears}, \quad \text{poverty} \longleftrightarrow \text{inability to heat adequately}.$$

This interpretation is consistent with the indicator-based definition of energy poverty, according to which a household is energy poor if it experiences at least one of the following conditions: inability to heat the dwelling adequately, arrears on utility bills, or the joint occurrence of high energy burden and poverty.

A further justification of (12)–(15) is methodological. If the model is required to rely only on quantities observed in EU-SILC or directly constructed from them, then it is preferable to avoid latent variables such as technical efficiency, physical energy need, or investment flows. The simplified system achieves this by operating directly on the observable deprivation dimensions themselves. It should therefore be understood as a parsimonious empirical abstraction rather than a fully structural model of household behavior.

In summary, the proposed representation is justified on three grounds. First, it respects the data constraint by using only EU-SILC-based dimensions. Second, it reflects the empirically plausible mutual reinforcement among heating inadequacy, arrears, energy burden, and poverty. Third, the decay terms ensure that the system has a meaningful interpretation as a dynamic process with both persistence and potential recovery.

5.3 Energy Poverty Indicators

In the following, we will use thresholding instead of concrete values because each country computes energy poverty using a different approach. Unlike our Czech threshold of 20% of income spent on energy, many European approaches do not rely on a single absolute cutoff. A widely used EU expenditure-based indicator is the “2M” measure, which identifies households whose energy expenditure share exceeds twice the national median, while some countries use fixed 10% rules and others rely on multidimensional indicator sets [?]. The binary indicators are then defined by thresholding:

$$I_1(t) = \mathbf{1}\{(h(t) \leq \tau_0) \vee (h(t) \geq \tau_1)\}, \quad (16)$$

$$I_2(t) = \mathbf{1}\{d(t) \geq \tau_2\}, \quad (17)$$

$$I_3(t) = \mathbf{1}\{b(t) \geq \tau_3 \wedge \underbrace{\text{INCOME}(t) \leq \tau_4}_{\substack{\text{financial poverty} \\ p(t)}}\}, \quad (18)$$

and the final energy-poverty status is

$$E_P(t) = \mathbf{1}\{I_1(t) \vee I_2(t) \vee I_3(t)\}.$$

5.4 Analytical Solutions

The simplified system can be written in matrix form as

$$\dot{x}(t) = Ax(t), \quad x(t) = \begin{bmatrix} h(t) \\ d(t) \\ b(t) \\ p(t) \end{bmatrix},$$

with

$$A = \begin{bmatrix} -\lambda_h & 0 & \alpha_{hb} & \alpha_{hp} \\ 0 & -\lambda_d & \alpha_{db} & \alpha_{dp} \\ 0 & \alpha_{bd} & -\lambda_b & \alpha_{bp} \\ \alpha_{ph} & \alpha_{pd} & 0 & -\lambda_p \end{bmatrix}.$$

Monotonicity. The simplified system belongs to the class of cooperative (monotone) dynamical systems. All off-diagonal entries of A are nonnegative, hence A is a Metzler matrix. Consequently, the flow generated by $\dot{x} = Ax$ is order-preserving: if $x(0) \leq y(0)$ componentwise, then $x(t) \leq y(t)$ for all $t \geq 0$.

This property implies that larger initial deprivation states cannot evolve into smaller ones faster than initially better-off states, reinforcing the interpretability of the dynamics in social terms. For the initial condition

$$x(0) = x_0 = \begin{bmatrix} h_0 \\ d_0 \\ b_0 \\ p_0 \end{bmatrix},$$

the unique solution is given by

$$x(t) = e^{At}x_0,$$

where e^{At} denotes the matrix exponential. The solution has an exponential form because the system is linear with constant coefficients. In the scalar case, the equation

$$\dot{x}(t) = \lambda x(t)$$

has the solution $x(t) = Ce^{\lambda t}$, since the exponential function is characterized by the property that its derivative is proportional to itself. In the vector case,

$$\dot{x}(t) = Ax(t),$$

The analogous object is the matrix exponential, and the solution is therefore given by

$$x(t) = e^{At}x_0.$$

Indeed, the matrix exponential satisfies

$$\frac{d}{dt}e^{At} = Ae^{At},$$

which implies

$$\frac{d}{dt}(e^{At}x_0) = Ae^{At}x_0 = Ax(t).$$

If A is diagonalizable, i.e.

$$A = V\Lambda V^{-1}, \quad \Lambda = \text{diag}(r_1, r_2, r_3, r_4),$$

then

$$x(t) = V e^{\Lambda t} V^{-1} x_0, \quad e^{\Lambda t} = \text{diag}(e^{r_1 t}, e^{r_2 t}, e^{r_3 t}, e^{r_4 t}).$$

Hence each component of the solution is a linear combination of exponential terms determined by the eigenvalues of A .

5.5 Equilibria of the Simplified System

To express the equilibrium of the simplified system in the form introduced in Section 5.4, we define an equilibrium $x^* = (h^*, d^*, b^*, p^*)^\top$ by the condition

$$Ax^* = 0,$$

that is,

$$0 = -\lambda_h h^* + \alpha_{hb} b^* + \alpha_{hp} p^*, \quad (19)$$

$$0 = -\lambda_d d^* + \alpha_{db} b^* + \alpha_{dp} p^*, \quad (20)$$

$$0 = -\lambda_b b^* + \alpha_{bd} d^* + \alpha_{bp} p^*, \quad (21)$$

$$0 = -\lambda_p p^* + \alpha_{pd} d^* + \alpha_{ph} h^*. \quad (22)$$

The trivial equilibrium

$$x^* = \mathbf{0}$$

always exists. Moreover, if $\det(A) \neq 0$, then this is the unique equilibrium. Nonzero equilibria exist only in the singular case $\det(A) = 0$, in which case the equilibrium set is the kernel of A .

The stability of the zero equilibrium is determined by the eigenvalues of A . In particular, $x^* = \mathbf{0}$ is asymptotically stable if all eigenvalues of A have negative real parts. A sufficient condition for this is strict diagonal dominance by rows, namely

$$\lambda_h > \alpha_{hb} + \alpha_{hp}, \quad (23)$$

$$\lambda_d > \alpha_{db} + \alpha_{dp}, \quad (24)$$

$$\lambda_b > \alpha_{bd} + \alpha_{bp}, \quad (25)$$

$$\lambda_p > \alpha_{pd} + \alpha_{ph}. \quad (26)$$

Under these inequalities, the decay effects dominate the reinforcing cross-effects, and all trajectories converge to the origin. It means that the self-damping effect in each equation is stronger than the total reinforcing influence of the remaining variables. Consequently, the feedback interactions are not strong enough to sustain a nonzero state, and all trajectories converge to the zero equilibrium. It translates into policy language as follows: energy poverty can be eradicated only if every mechanism that generates it is counteracted more strongly than it is reproduced by the rest of the social-energy system.

In practical terms, the stability inequalities mean that a society can eliminate energy poverty only if the direct corrective mechanisms in each dimension are stronger than the reinforcing spillovers coming from the remaining dimensions. Thus, policies must not only reduce individual manifestations of energy poverty, such as inadequate heating, arrears, or excessive energy burdens, but also weaken the feedback channels through which these problems reproduce one another. A stable zero equilibrium, therefore, corresponds to a policy regime in which housing efficiency, income protection, debt management, and affordability measures jointly dominate the systemic forces generating energy deprivation.

5.6 Illustrative Dynamic Simulations

All codes used in the simulations below are available in the repository <https://github.com/ZahraAlijani/energy-poverty>; an interactive version is also provided at <https://colab.research.google.com/drive/1BPsg4uu6w58Y-TCjLL7HY42we5hLK4HZ?usp=sharing>. All figures of this paper are produced from the model itself by the script `paper_figures.py`, which integrates the system, evaluates the fuzzy indicators, and reports the eigenvalues and index levels quoted in the text; see Appendix A and Table 3 for the correspondence between the figures and the code.

While the analytical results characterize stability and persistence in terms of eigenvalues and parameter inequalities, it is useful to illustrate the qualitative implications of these results by means of numerical simulations. The simulations presented below are intended to demonstrate typical dynamic patterns of the simplified system under different parameter regimes rather than to provide empirical calibration or forecasting.

5.6.1 Poverty-Trap Regime without Direct Energy-Burden Damping

We first consider a parameter configuration in which heating inadequacy, arrears, and poverty exhibit intrinsic recovery, while the energy-burden dimension lacks direct self-damping. Formally, this corresponds to $\lambda_h > 0$, $\lambda_d > 0$, $\lambda_p > 0$, and $\lambda_b = 0$.

This configuration represents a situation in which affordability pressures are not structurally corrected, but persist through reinforcing feedback from arrears and poverty. Such a regime may reflect policy environments relying primarily on temporary subsidies or income transfers without addressing the underlying drivers of high relative energy costs.

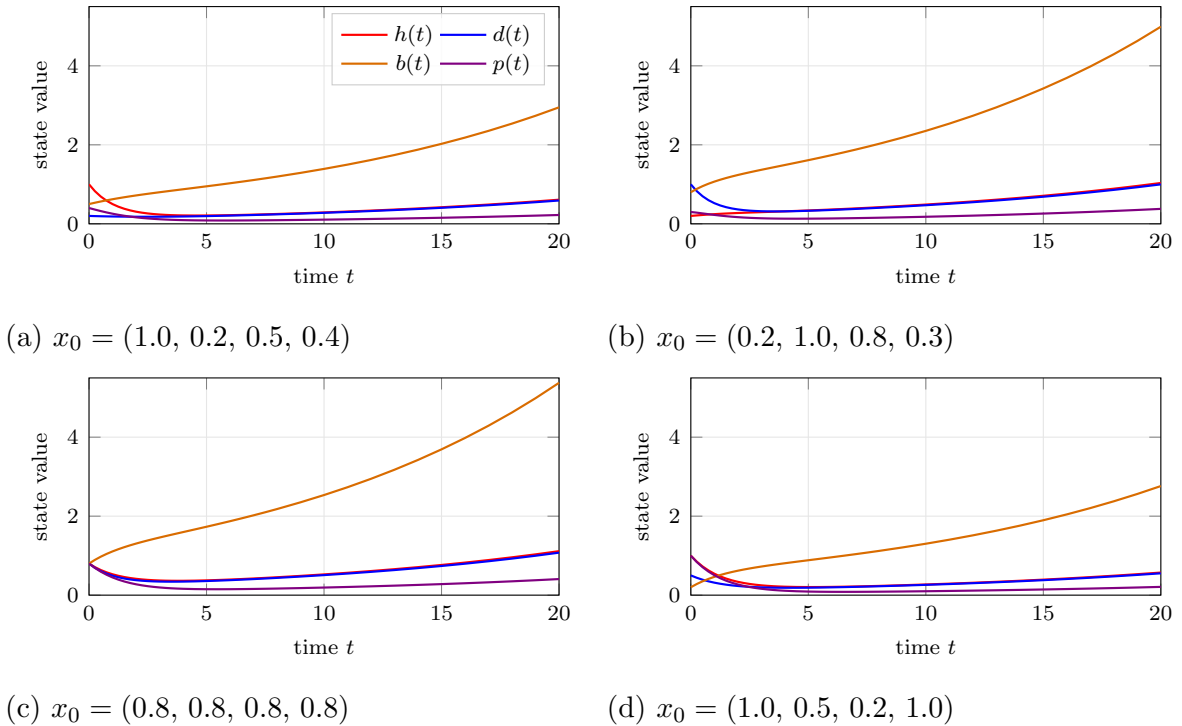


Figure 1: Multi-panel simulation of the simplified energy poverty dynamics in a poverty-trap regime ($\lambda_b = 0$, $\lambda_h = \lambda_d = \lambda_p = 1$). Each panel corresponds to a different initial condition and shows the time evolution of inability to heat the dwelling $h(t)$, arrears $d(t)$, energy burden $b(t)$, and poverty $p(t)$. Since $\lambda_b = 0$, the invariance of $[0, 1]^4$ is not guaranteed in this regime; values of the energy burden above one are plotted as generated by the linear system and correspond, under the intensity interpretation, to a state truncated at the upper boundary.

The eigenvalues of the system matrix in this regime consist of three negative values and one small positive dominant eigenvalue, numerically about 0.075 for the parameters used in Figure 1. Consequently, the zero equilibrium is unstable. Although some deprivation dimensions initially decline due to positive self-damping parameters, the absence of direct damping in the energy-burden equation allows reinforcing feedback effects to dominate in the long run.

The resulting dynamics exhibit slow but persistent deprivation rather than rapid divergence. This behavior corresponds to a dynamic poverty trap: energy poverty does not disappear endogenously, even in the absence of new shocks, and small disturbances may lead to long-lasting affordability stress.

5.6.2 Stable Regime with Direct Energy-Burden Damping

We next consider a modified parameter configuration in which the energy-burden dimension is endowed with a positive self-damping term, $\lambda_b > 0$, while all other parameters remain unchanged. This setting represents a policy environment in which affordability pressures are structurally addressed, for example through lasting reductions in energy intensity, regulated tariffs, or efficiency improvements.

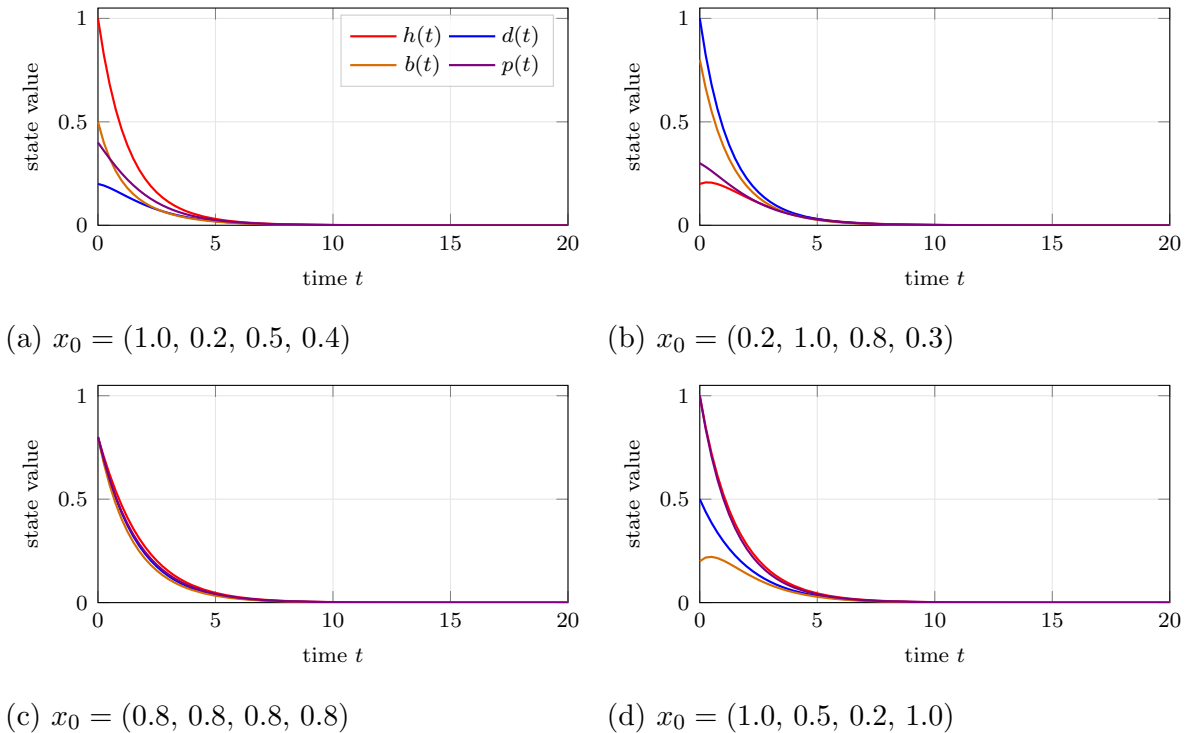


Figure 2: Multi-panel simulation of the simplified energy poverty dynamics in a stable regime ($\lambda_b = 1.2 > 0$, all other parameters unchanged). All four state variables converge to zero independently of initial conditions, indicating global asymptotic stability.

In this regime, all eigenvalues of the system matrix have strictly negative real parts; the dominant one is about -0.598 for the parameters used in Figure 2. The zero equilibrium is therefore globally asymptotically stable, and all dimensions of deprivation converge to zero over time, as the four panels show. Short-run oscillations may occur due to complex eigenvalues, but they are damped and do not generate persistent effects.

The qualitative difference between Figures 1 and 2 highlights the critical role of direct energy-burden damping. Once reinforcing feedback loops are counteracted by sufficiently strong corrective mechanisms, the system no longer sustains energy poverty endogenously.

From a policy perspective, the comparison between the two regimes illustrates that reducing income poverty or arrears alone may be insufficient to eliminate energy poverty. Without direct interventions targeting energy affordability, feedback mechanisms may continue to reproduce deprivation. In contrast, policies that structurally reduce energy burden break the feedback loop and ensure convergence toward the non-deprived state.

6 Fuzzy Representation of Deprivation Dimensions

The simplified system evolves on continuous deprivation intensities $h(t)$, $d(t)$, $b(t)$, and $p(t)$. To connect these states with indicator-based EU-SILC definitions, we introduce fuzzy membership functions. The operators, transitions and indicators of this section are implemented in the module `FuzzyEnergyPoverty.py` of the repository referenced in Appendix A; all figures below are generated by that implementation.

Fuzzy operators. We use the product t-norm

$$x \otimes y = xy$$

and the probabilistic sum

$$x \oplus y = x + y - xy.$$

Smooth thresholds. For $L < U$, we define cosine transitions

$$\Psi_{0 \rightarrow 1}(x; L, U) = \begin{cases} 0, & x \leq L, \\ \frac{1 - \cos\left(\pi \frac{x - L}{U - L}\right)}{2}, & L < x < U, \\ 1, & x \geq U, \end{cases} \quad (27)$$

and analogously $\Psi_{1 \rightarrow 0}(x; L, U)$. Both functions are C^1 -smooth and have a zero slope at L and U . Since the above transitions are intended to express gradual inequality, we will use relational notation for these mappings, as they are membership functions of binary fuzzy relations:

$$(x \succeq_L U) = \Psi_{0 \rightarrow 1}(x; L, U) \quad \text{and} \quad (x \preceq_L U) = \Psi_{1 \rightarrow 0}(x; L, U) \quad (28)$$

6.1 Fuzzy Energy Poverty Indicators

Each binary indicator in (16) is obtained by comparing a state variable with a crisp threshold. Replacing every such comparison by the corresponding gradual relation (28), conjunction by the t-norm $\otimes$ and disjunction by the t-conorm $\oplus$, we define

$$\tilde{I}_1(t) = (h(t) \preceq_{L_H} U_H) \oplus (h(t) \succeq_{L_C} U_C), \quad \text{heating and cooling indicator,} \quad (29)$$

$$\tilde{I}_2(t) = (d(t) \succeq_{L_d} U_d), \quad \text{debt indicator,} \quad (30)$$

$$\tilde{I}_3(t) = (b(t) \succeq_{b_L} b_U) \otimes \underbrace{(y(t) \preceq_{y_L} y_U)}_{\mu_p(t)}, \quad \text{energy expenses and income indicator,} \quad (31)$$

where $y(t)$ denotes the income variable entering the poverty condition of (16) and $\mu_p(t) \in [0, 1]$ is the resulting degree of financial poverty.

In each case, the pair of thresholds brackets the crisp cut-off that it replaces,

$$L_H < \tau_0 < U_H, \quad L_C < \tau_1 < U_C, \quad L_d < \tau_2 < U_d, \quad b_L < \tau_3 < b_U, \quad y_L < \tau_4 < y_U,$$

and the transition width $\Delta = U - L$ measures how much gradedness the definition admits. If every transition is centered at the corresponding crisp threshold, that is $\tau = (L + U)/2$, then the symmetry of the cosine transition gives $\Psi_{0 \rightarrow 1}(\tau; L, U) = \frac{1}{2}$, so that thresholding a fuzzy indicator at the level $\frac{1}{2}$ reproduces exactly the original binary classification. Letting $\Delta \rightarrow 0$ in all transitions, $\Psi_{0 \rightarrow 1}(x; L, U) \rightarrow \mathbf{1}\{x \geq \tau\}$ for every $x \neq \tau$, the product t-norm reduces to conjunction and the probabilistic sum to disjunction. The binary indicators (16) are therefore recovered as a limiting case, and the fuzzy formulation is a proper generalization rather than a different measurement concept.

The cosine transitions are particularly suitable for this purpose because they are C^1 -smooth and have zero derivatives at both endpoints. Hence, the indexes change continuously without abrupt jumps or sharp changes in slope when a state variable crosses one of the thresholds.

State-space form of the poverty membership. The poverty component of the dynamical system is the intensity $p(t) \in [0, 1]$, in which higher values correspond to more severe deprivation, whereas $y(t)$ in (31) is an income variable, in which higher values correspond to less deprivation. Whenever the indicators are evaluated directly along the trajectories of the dynamical system, the same membership degree is therefore expressed as

$$\mu_p(t) = (p(t) \succeq_{p_L} p_U), \quad (32)$$

with $p_L < p_U$ bracketing the intensity level at which the household is regarded as financially poor. Both variants are used below: the income form is convenient for interpretation in EU-SILC units, whereas (32) keeps all three indicators as functions of the state vector $x(t) = (h, d, b, p)^\top$ alone.

Thermal inadequacy as an intensity. The same distinction applies to the first indicator. If $h(t)$ is measured as the indoor temperature, the two-sided form (29) is required, because deprivation occurs at both ends of the temperature scale. If, as in the simplified dynamical system, $h(t)$ is the intensity of thermal inadequacy, which already aggregates insufficient heating and insufficient cooling into a single deprivation-increasing variable, the indicator reduces to the one-sided form

$$\tilde{I}_1(t) = (h(t) \succeq_{L_h} U_h). \quad (33)$$

The two readings are used consistently below: the temperature form in Section 6.1.1, where the construction is illustrated in physical units, and the intensity form (33) whenever the indicators are evaluated along trajectories of the dynamical system.

6.1.1 Heating and Cooling Indicator

Figure 3 illustrates the construction of the fuzzy heating-and-cooling indicator given by (29), where $h(t)$ denotes the indoor temperature at time t , the pair $L_H < U_H$ delimits the transition associated with heating demand, and the pair $L_C < U_C$ the transition associated with cooling demand. The first component describes the degree to which the indoor temperature is sufficiently low to indicate a need for heating, whereas the second component describes the degree to which it is sufficiently high to indicate a need for cooling.

The left panel shows the heating component. Two thresholds, denoted by L_H and U_H , determine the transition interval. For temperatures $h(t) \leq L_H$, the membership degree is equal to one, indicating a fully active heating demand. For temperatures $h(t) \geq U_H$, the membership degree is zero, indicating that heating is not required. Between these thresholds, the membership degree decreases smoothly according to the cosine transition (28)

The right panel represents the cooling component. Its transition is controlled by the thresholds L_C and U_C . For temperatures $h(t) \leq L_C$, the cooling membership degree is zero, since the indoor temperature is not sufficiently high to require cooling. For temperatures $h(t) \geq U_C$, the

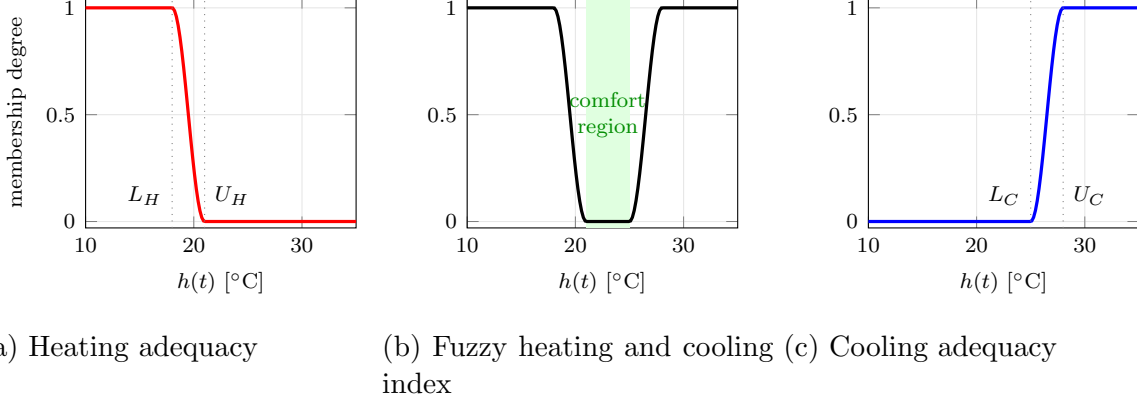


Figure 3: Heating adequacy, the resulting fuzzy index, and cooling adequacy as functions of indoor temperature.

membership degree is one, corresponding to a fully active cooling demand. Within the transition interval (L_C, U_C) , the cooling degree increases smoothly according to $\Psi_{0 \rightarrow 1}(h(t); L_C, U_C)$ (28)

The central panel shows the resulting fuzzy index obtained by aggregating the two components:

$$\tilde{I}_1(t) = \mu_H(h(t)) \oplus \mu_C(h(t)).$$

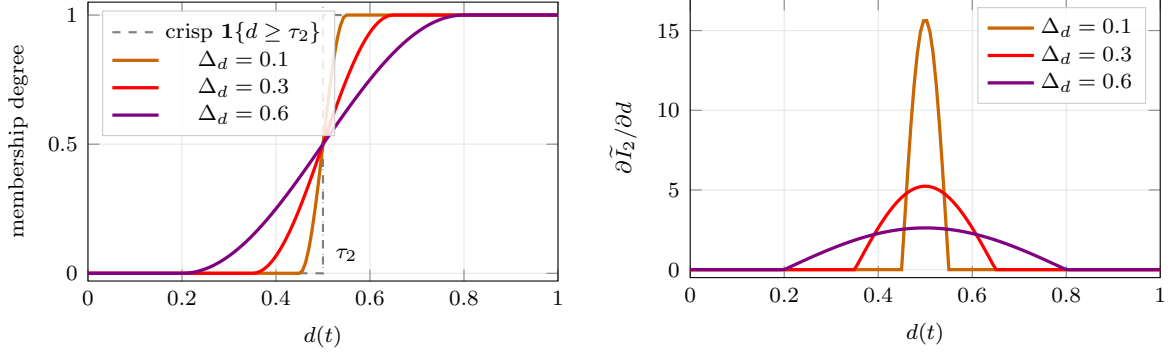
High values of $\tilde{I}_1(t)$ therefore indicate that either heating or cooling is needed. Conversely, low values correspond to the thermally adequate region in which neither component is substantially active.

The thresholds have a direct practical interpretation. The values L_H and U_H determine how gradually the assessment changes from a clear heating requirement to thermal adequacy. Similarly, L_C and U_C determine the transition from thermal adequacy to a clear cooling requirement. Increasing the width $U_H - L_H$ or $U_C - L_C$ produces a more gradual transition, whereas decreasing it makes the corresponding criterion closer to a crisp temperature threshold. In the illustrated specification, the cooling transition is placed at higher temperatures than the heating transition, creating an intermediate comfort region in which both membership degrees are close to zero.

6.1.2 Debt Indicator

The debt indicator (30) replaces the crisp arrears condition $\mathbf{1}\{d(t) \geq \tau_2\}$ by the gradual relation $(d(t) \succeq_{L_d} U_d)$. In EU-SILC, arrears are recorded as a binary answer to the question whether the household was unable to pay utility bills on time during the preceding twelve months. The corresponding state variable $d(t)$ of the dynamical system is, however, a continuous measure of accumulated payment difficulty. Depending on the level of aggregation, it may be interpreted as the amount of outstanding energy debt relative to monthly income, as the length of the payment delay, or as the prevalence of arrears within a population group. Below L_d , payment delays are occasional and do not indicate financial distress; above U_d , the household is unambiguously in arrears and exposed to enforcement or disconnection. Between the two thresholds, $\tilde{I}_2(t)$ expresses how far the process of indebtedness has advanced.

Figure 4(a) shows the indicator for three transition widths centered at the same crisp threshold. All three curves pass through the value $\frac{1}{2}$ at $d(t) = \tau_2$, so they classify households identically once a decision at level $\frac{1}{2}$ is required, while differing in how the intermediate region is described. Narrowing the interval moves the fuzzy indicator toward the binary step function, which is recovered in the limit $\Delta_d \rightarrow 0$.



(a) Debt indicator $\tilde{I}_2$

(b) Sensitivity $\partial\tilde{I}_2/\partial d$

Figure 4: Fuzzy debt indicator and its sensitivity for three transition widths $\Delta_d = U_d - L_d$ centered at the crisp threshold $\tau_2 = 0.5$. The dashed line in the left panel is the binary indicator obtained in the limit $\Delta_d \rightarrow 0$.

The role of the transition width becomes explicit in the sensitivity of the indicator. Differentiating the cosine transition (27) gives

$$\frac{\partial\tilde{I}_2}{\partial d} = \Psi'_{0 \rightarrow 1}(d; L_d, U_d) = \begin{cases} \frac{\pi}{2\Delta_d} \sin\left(\pi \frac{d - L_d}{\Delta_d}\right), & L_d < d < U_d, \\ 0, & \text{otherwise,} \end{cases} \quad (34)$$

which vanishes at both endpoints and attains its maximum $\pi/(2\Delta_d)$ at the midpoint, as illustrated in Figure 4(b). Consequently, a narrow transition concentrates the entire responsiveness of the indicator into a small neighborhood of the threshold and reproduces the instability of binary classification, in which negligible changes in arrears reverse the assessment. A wider transition distributes the responsiveness over a larger range of states and thus provides an early-warning signal before the administrative threshold is reached, at the cost of a weaker correspondence with the legal or statistical meaning of τ_2 . In practice, Δ_d should be chosen so that it reflects the measurement uncertainty of $d(t)$ rather than being set arbitrarily.

Because $\tilde{I}_2$ is a smooth function of a single state variable, its evolution along a trajectory of the simplified system follows from the chain rule and (13),

$$\frac{d}{dt}\tilde{I}_2(t) = \Psi'_{0 \rightarrow 1}(d(t); L_d, U_d) \dot{d}(t) = \Psi'_{0 \rightarrow 1}(d(t); L_d, U_d) (-\lambda_d d(t) + \alpha_{db} b(t) + \alpha_{dp} p(t)). \quad (35)$$

The measured degree of deprivation therefore changes only for households whose arrears lie inside the transition band, and its rate of change is the product of a purely measurement-related factor and the dynamics generated by energy burden and poverty.

6.1.3 Energy Expenses and Income Indicator

The third indicator differs from the previous two in that it is conjunctive: according to (16), a household is deprived in this dimension only if a high share of energy expenditure occurs *simultaneously* with low income. Its fuzzy counterpart (31) is accordingly the t-norm of two membership degrees,

$$\tilde{I}_3(t) = \mu_b(t) \otimes \mu_p(t) = \mu_b(t) \mu_p(t), \quad \mu_b(t) = (b(t) \succeq_{b_L} b_U), \quad \mu_p(t) = (y(t) \preceq_{y_L} y_U).$$

Figure 5(a) shows the burden component. The thresholds b_L and b_U are placed symmetrically around the national cut-off, here the Czech value $\tau_3 = 0.20$ of the expenditure share; in another

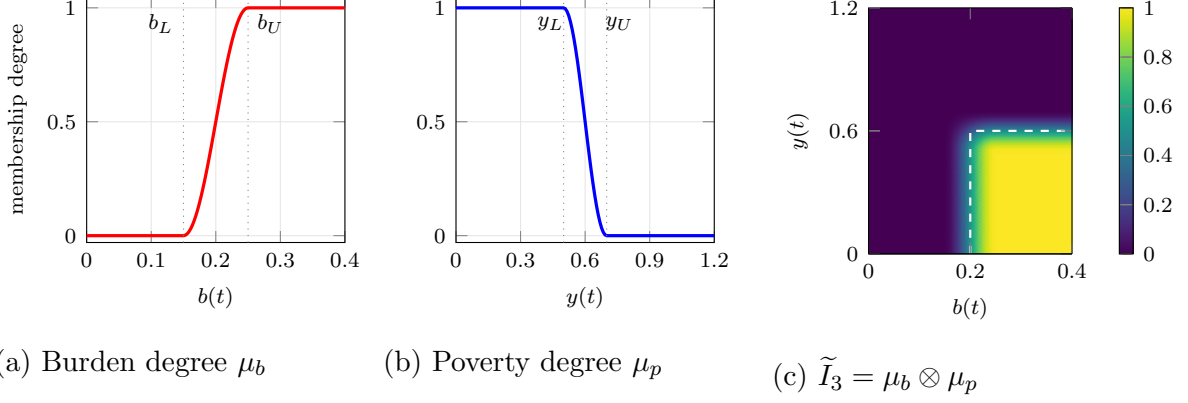


Figure 5: Construction of the fuzzy energy-expenses-and-income indicator with $b_L = 0.15$, $b_U = 0.25$ bracketing the burden threshold $\tau_3 = 0.20$, and $y_L = 0.5$, $y_U = 0.7$ bracketing the income threshold $\tau_4 = 0.6$ of the median residual income. The dashed line in panel (c) delimits the crisp deprivation region $\{b \geq \tau_3\} \cap \{y \leq \tau_4\}$ replaced by the graded one.

operationalization, the same construction applies with τ_3 given by twice the national median share used in the “2M” indicator, or by any other convention adopted in the country under study. Figure 5(b) shows the income component, in which the thresholds bracket the poverty line $\tau_4 = 0.6$ of the median residual income. Since low income is the deprivation-increasing direction, the corresponding membership is decreasing.

Figure 5(c) displays the resulting indicator on the plane of the two arguments. The dashed line indicates the boundary of the crisp deprivation region that the fuzzy indicator replaces. The graded formulation smooths this boundary in both directions simultaneously, so that households situated near the corner (τ_3, τ_4) , which the binary definition separates sharply, receive intermediate degrees. Four properties of this construction are worth emphasizing.

- (P1) **Consistency with the crisp definition.** $\tilde{I}_3(t) = 1$ if and only if $b(t) \geq b_U$ and $y(t) \leq y_L$, and $\tilde{I}_3(t) = 0$ if and only if $b(t) \leq b_L$ or $y(t) \geq y_U$. The set of fully deprived households is thus an inner core of the crisp region, the set of positive degrees is an outer envelope of it, and the two coincide as the transition widths tend to zero.
- (P2) **Monotonicity.** $\tilde{I}_3$ is nondecreasing in the energy burden and nonincreasing in income. Hence, no improvement in either dimension can increase the measured deprivation, which is a necessary requirement for an indicator used in policy evaluation.
- (P3) **Absence of compensation.** In contrast to the disjunctive aggregation used in $\tilde{I}_1$, a high energy burden alone cannot generate deprivation in this dimension if income is adequate, and vice versa. This is precisely the intent of the joint condition in the EU–SILC definition. Moreover, since $xy \leq \min\{x, y\}$, the product t-norm yields a conservative assessment: the degree never exceeds the weaker of the two components.
- (P4) **Smoothness.** The product of two C^1 transitions is C^1 on the whole plane. The minimum t-norm $x \wedge y$ would also satisfy (P1)–(P3), but it introduces a kink along the set $\mu_b = \mu_p$ and is therefore not differentiable there, which would prevent the sensitivity analysis of the composite index below. The product is in addition strictly increasing in both arguments whenever both are positive, so that improvements in either dimension are always registered.

The sensitivity of the indicator follows from the product rule. Using the state-space form (32) of the poverty membership and the derivative (34) of the cosine transition,

$$\frac{\partial \tilde{I}_3}{\partial b} = \mu_p(t) \Psi'_{0 \rightarrow 1}(b(t); b_L, b_U), \quad \frac{\partial \tilde{I}_3}{\partial p} = \mu_b(t) \Psi'_{0 \rightarrow 1}(p(t); p_L, p_U), \quad (36)$$

and consequently, by (14) and (15),

$$\begin{aligned} \frac{d}{dt} \tilde{I}_3(t) = & \mu_p(t) \Psi'_{0 \rightarrow 1}(b(t); b_L, b_U) (-\lambda_b b(t) + \alpha_{bd} d(t) + \alpha_{bp} p(t)) \\ & + \mu_b(t) \Psi'_{0 \rightarrow 1}(p(t); p_L, p_U) (-\lambda_p p(t) + \alpha_{pd} d(t) + \alpha_{ph} h(t)). \end{aligned} \quad (37)$$

The multiplicative structure of (36) has a direct substantive meaning. A change in the energy burden affects the measured deprivation only to the extent that the household is already financially poor, and a change in poverty matters only to the extent that the energy burden is already high. The indicator therefore responds to the coincidence of the two pressures rather than to either of them in isolation, and it is most sensitive for households that are simultaneously in both transition bands. This is the population that binary indicators classify least reliably and that is of primary interest for early intervention.

6.2 Fuzzy Energy Poverty Index

Let $w_1 + w_2 + w_3 = 1$, $w_i \geq 0$. The fuzzy energy poverty index is defined as

$$\tilde{P}(t) = w_1 \tilde{I}_1(t) \oplus w_2 \tilde{I}_2(t) \oplus w_3 \tilde{I}_3(t). \quad (38)$$

It is the gradual counterpart of the binary status $E_P(t) = \mathbf{1}\{I_1 \vee I_2 \vee I_3\}$, the disjunction being replaced by the t-conorm $\oplus$. Since $\oplus$ is monotone and $[0, 1]$ is closed under it, $\tilde{P}(t) \in [0, 1]$, and $\tilde{P}(t)$ increases whenever any of the three dimensions deteriorates. Note that the weights rescale the individual dimensions before aggregation, so that under equal weights $w_i = \frac{1}{3}$ and full deprivation in all three dimensions the index attains $\frac{1}{3} \oplus \frac{1}{3} \oplus \frac{1}{3} \approx 0.704$ rather than one. The index should therefore be read as an ordinal measure of systemic stress; if a normalized scale is required, the weights must be chosen so that $\bigoplus_i w_i = 1$.

6.2.1 Behavior of the Index Along Trajectories

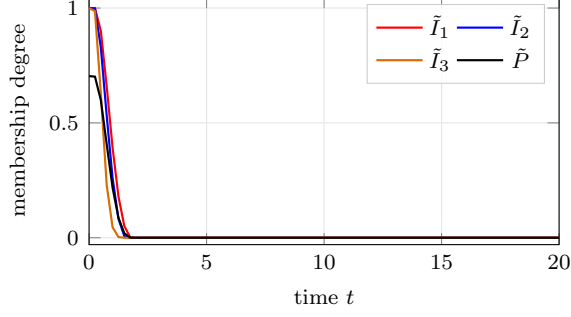
Because each fuzzy indicator is a smooth function of the state, the index evaluated along a solution of the simplified system, $\tilde{P}(t) = \tilde{P}(x(t))$, is differentiable and satisfies

$$\frac{d}{dt} \tilde{P}(t) = \nabla \tilde{P}(x(t)) \cdot Ax(t).$$

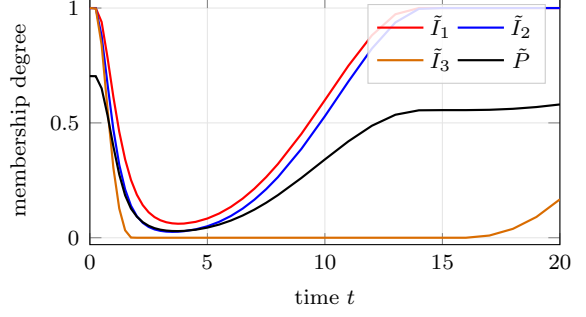
It therefore inherits the qualitative behavior established in Section 5.4: if all eigenvalues of A have negative real parts, then $x(t) \rightarrow 0$ and, since $\tilde{P}(0) = 0$ at the origin, $\tilde{P}(t) \rightarrow 0$; if A has an eigenvalue with positive real part, the state does not converge to the origin and $\tilde{P}(t)$ remains bounded away from zero.

Figure 6 illustrates both cases for the trajectory starting from full deprivation in all four dimensions. In the stable regime, Figure 6(a), the indicators leave their saturation level in a definite order. The conjunctive indicator $\tilde{I}_3$ vanishes first, since it is a product of two memberships that both decrease, followed by the arrears indicator and finally by the thermal one; the whole index reaches zero within approximately two time units. Deprivation is thus eliminated endogenously once the energy burden is damped directly.

In the poverty-trap regime, Figure 6(b), the picture is qualitatively different. The index first declines steeply, to $\tilde{P} \approx 0.04$ around $t = 3$, because the dimensions with positive self-damping recover quickly. The decline then stops and reverses, as the undamped energy burden feeds back through arrears and poverty. The dominant eigenvalue of the system matrix in this regime is approximately 0.075, so deprivation grows slowly rather than explosively; the relapse appears first in the thermal and arrears components, which return to saturation by the end of the horizon, while $\tilde{I}_3$ begins to rise only at its very end. At $t = 20$ the index has recovered to $\tilde{P} \approx 0.58$ and keeps increasing. The comparison shows the practical value of a gradual measure: around $t = 3$



(a) Stable regime ($\lambda_b = 1.2$)



(b) Poverty-trap regime ($\lambda_b = 0$)

Figure 6: Fuzzy indicators and the composite index (38) along the trajectory starting from $x_0 = (0.8, 0.8, 0.8, 0.8)$ in the two regimes of Figures 1 and 2, with equal weights $w_i = \frac{1}{3}$ and transitions $[0.3, 0.7]$ in every dimension.

the two regimes are nearly indistinguishable in terms of a binary classification, since almost no household would be recorded as energy poor, whereas the trajectory of $\tilde{P}$ separates a system that has been stabilized from one that is only temporarily improving before a relapse.

Because $\tilde{P}(t)$ is continuous and sensitive to intermediate levels of vulnerability, it can be used as an early-warning statistic rather than only as an ex-post classification. Its derivative is largest for households situated inside the transition bands, that is, precisely for those whose binary status is least stable.

A Reproducibility and Source Code

Every figure in this paper is generated from the model itself rather than drawn by hand, and the numerical values quoted in the text can be recomputed from the same source. The complete implementation is openly available in the repository

<https://github.com/ZahraAlijani/energy-poverty>

and consists of two Python modules. The module `FuzzyEnergyPoverty.py` implements the fuzzy layer of Section 6: the product t-norm, the probabilistic sum, the cosine transitions (27) together with their derivative (34), the three indicators, and the composite index (38). The script `paper_figures.py` integrates the simplified system (12)–(15) and draws all figures, collecting every parameter that appears in a caption in a single configuration block; running it reproduces the figures of this paper together with the eigenvalues and index levels quoted in the text. Table 3 maps the figures to the functions that produce them.

Output	Section	Function in <code>paper_figures.py</code>
Figure 1	5.6.1	<code>figure_trajectories("trap", ...)</code>
Figure 2	5.6.2	<code>figure_trajectories("stable", ...)</code>
Figure 3	6.1.1	<code>figure_heating_cooling()</code>
Figure 4	6.1.2	<code>figure_debt()</code>
Figure 5	6.1.3	<code>figure_expenses_income()</code>
Figure 6	6.2.1	<code>figure_fuzzy_index()</code>
Eigenvalues and index levels	5.6, 6.2.1	<code>report()</code>

Table 3: Correspondence between the figures of the paper and the code.

The repository also contains `TrajectoriesSimulation.py`, an earlier standalone version of the simulation, and the notebook used for exploratory runs. Only `numpy`, `scipy` and `matplotlib` are required, so the figures can be regenerated with

```
python paper_figures.py
```

either locally or in a hosted notebook environment.